

Demonstration of Proposed Features

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Demonstration of Proposed Features:

S.No	Feature Name	Description	Status	Demonstrated (Yes/No)	Remarks
1	Dataset Loading	Load HDI.csv (195 rows x 82 cols) into Pandas DataFrame	Implemented	Yes	Shown in Epic 3 notebook
2	EDA - Strip Plots	Strip plot: Life Expectancy vs HDI, Schooling vs HDI	Implemented	Yes	Plots saved in plots/ folder
3	EDA - Heatmap	Correlation heatmap of key numeric features	Implemented	Yes	Clearly shows HDI correlations
4	Data Preprocessing	Null handling + Label Encoding of Country column	Implemented	Yes	X.isnull().sum() demonstrated
5	Train/Test Split	90% train (175) / 10% test (20) split	Implemented	Yes	Shown in Epic 5 notebook
6	Model Training	Linear Regression trained on x_train	Implemented	Yes	reg.fit() demonstrated
7	Model Evaluation	R-squared = 93.74%, Actual vs Predicted plot	Implemented	Yes	Scatter plot generated
8	Model Saving	HDI.pkl saved via Pickle	Implemented	Yes	File verified in model/ folder
9	Flask Home Page	home.html with HDI description and navigation	Implemented	Yes	Browser demo at localhost:5000
10	Prediction Form	Country dropdown + 4 input fields	Implemented	Yes	Live form submission tested
11	HDI Prediction	Real-time score + category output	Implemented	Yes	Multiple countries tested
12	Result Display	resultnew.html shows prediction in gold text	Implemented	Yes	'Medium HDI 0.48' displayed

13	Navigation	All 3 routes functional (/,Prediction,/predict)	Implemented	Yes	Full navigation flow shown
14	Category Classification	Low/Medium/High/Very High categories	Implemented	Yes	All 4 categories tested

Feature Implementation Summary:

Metric	Value
Total Features Proposed	14
Total Features Implemented	14
Total Features Demonstrated	14
Overall Implementation Rate (%)	100%